THE STALK OF THE PUSHFORWARD ALONG A FINITE MORPHISM IS THE PRODUCT OF THE STALKS OVER THE GEOMETRIC LIFTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The stalk of a pushforward computed after an etale base change. Properties of Spaces, label spaces-properties-lemma-pushforward-etale-base-change (invoked at proof line 535). It lets the computation be performed after replacing Y by an etale neighbourhood of the geometric point, and X by the corresponding fibre product.
- The statement for a finite morphism of schemes. Etale Cohomology, proposition etale-cohomology-proposition-finite-higher-direct-image-zero (invoked at proof line 538). After the reduction to schemes it identifies the stalk of the pushforward with the product of the stalks at the finitely many lifts.

Source reference. Original proposition 03QP: sources/stacks-chow/tags.txt, source lines 4902–4902. Navigation pointer for \ref{etale-cohomology-proposition-finite-higher-direct-image-zero} used inside the counted span (line 539); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LR: sources/stacks-chow/tags.txt, source lines 4723–4723. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change} used inside the counted span (line 535); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a finite morphism of algebraic spaces over S. Let $\overline{y}$ be a geometric point of Y with lifts $\overline{x}_1, \ldots, \overline{x}_n$ in X. Then*

$$(f_*\mathcal{F})_{\overline{y}} = \prod\nolimits_{i=1,\ldots,n} \mathcal{F}_{\overline{x}_i}$$

for any sheaf $\mathcal{F}$ on $X_{étale}$.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose an étale neighbourhood $(V, \overline{v})$ of $\overline{y}$. Then the stalk $(f_*\mathcal{F})_{\overline{y}}$ is the stalk of $f_*\mathcal{F}|_V$ at $\overline{v}$. By Properties of Spaces, Lemma 03LR we may replace Y by V and X by $X \times_Y V$. Then $Z \to X$ is a finite morphism of schemes and the result is Étale Cohomology, Proposition 03QP. □